"Tracy Fullerton's *Game Design Workshop* covers pretty much everything a working or wannabe game designer needs to know. She covers game theory, concepting, prototyping, testing and tuning, with stops along the way to discuss what it means to be a professional game designer and how to land a job. When I started thinking about my game studies course at the University of Texas at Austin, this was one book I knew I had to use."

—Warren Spector, *Creative Director, Junction Point Studios*

"This is a break-through book, brimming with battle-tested 'how-to's.' Aspiring game designers: you will 'break through' to the next level when you learn to set, and then test, experience goals for your game players."

—Bing Gordon, *Chief Creative Officer, Electronic Arts*

"*Game Design Workshop* is without a question the most important (and best book) on the topic of game design. Its unique approach is both deep and practical and draws students' into the very heart of what game design is all about. The emphasis on paper-and-pencil prototyping encourages students to think, quite literally, "outside the box," and stretch themselves to innovate beyond simply rehashing commercially successful game genres. If the author's students are any indication, this method has a proven track record of producing both original and successful games. *Game Design Workshop* is ideal for those starting new educational programs as the book is structured around a design curriculum that can be easily implemented by instructors with no prior game design experience."

—Celia Pearce, *Director, Experimental Game Lab, Georgia Institute of Technology*

"This book offers a thoughtful and comprehensive look at the field of game design. I'm particularly impressed with the way Tracy has managed to integrate the viewpoints and comments of so many diverse and notable designers with her own perceptive view of the state of the art."

—Noah Falstein, *freelance designer, The Inspiracy*

"With the second edition of *Game Design Workshop*, the authors have kept the engaging hand-on exercise-based approach, while giving the text a nice updating and polishing. This book does a great job illustrating how games are designed and developed by engaging readers to play along."

—Drew Davidson, *Director, Entertainment Technology Center, Carnegie Mellon University*

"If you are considering becoming a professional game designer, you will find this book a reliable, intelligent, and compassionate guide. If you are already a professional game designer, you'll find this book an inspiration."

—Bernie DeKoven, *deepfun.com*

"If you have ever wanted to design a video game, *Game Design Workshop* is the book you want."

—Jesper Juul, *video game theorist and designer, author of* Half-Real

"Tracy Fullerton has combined her innate understanding and joy of games with her patient and objective experience as a scholar in this excellent book. She'll make you a better game developer with her clever exercises and concise prose. This is a must-have in the library of anyone serious about their games."

—John Hight, *Director of External Production, Sony Computer Entertainment of America*

"Game design is something of a black art. The trick to doing it well is retaining the black magic but training oneself to control it. There are a lot of books on game design out there, but *Game Design Workshop* is among the very few that develops a wizard rather than a drone."

—Ian Bogost, *professor of digital media, the Georgia Institute of Technology, and Co-Founder, Persuasive Games*

GAME DESIGN WORKSHOP

GAME DESIGN WORKSHOP

A Playcentric Approach to Creating Innovative Games

Second Edition

Tracy Fullerton

with Christopher Swain and Steven S. Hoffman

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SAN FRANCISCO • SINGAPORE • SYDNEY • TOKYO
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Composition diacriTech
Copyeditor Jeanne Hansen
Proofreader Troy Lilly
Indexer Michael Ferreira
Interior printer Sheridan Books, Inc.
Cover printer Phoenix Color, Inc.

Morgan Kaufmann Publishers is an imprint of Elsevier.
30 Corporate Drive, Suite 400, Burlington, MA 01803, USA

This book is printed on acid-free paper. ∞

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Library of Congress Cataloging-in-Publication Data
Fullerton, Tracy.
Game design workshop : a playcentric approach to creating innovative games / Tracy Fullerton, with Christopher Swain, and Steven S. Hoffman. —2nd ed.
p. cm.
Includes bibliographical references and index.
ISBN 978-0-240-80974-8 (pbk. : alk. paper) 1. Computer games—Programming. 2. Computer games—Design. 3. Computer graphics. I. Swain, Christopher, 1966–II. Hoffman, Steven, 1965–III. Title.

QA76.76.C672F84 2008
794.8'1526—dc22 2007040857

ISBN: 978-0-240-80974-8

For information on all Morgan Kaufmann publications,
visit our Web site at *www.mkp.com* or *www.books.elsevier.com*

Printed in the United States.
08 09 10 11 5 4 3 2 1

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ELSEVIER BOOK AID International Sabre Foundation

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Foreword

Eric Zimmerman, Co-Founder & Chief Design Officer, Gamelab

> *There is a connection. Every point in my life is connected to every other point. The connection is there. One need only imagine in full freedom.*
> *— Peter Handke*

There is magic in games.

Not magic like a Level 19 fireball spell is magic. Not the kind of magic you get when you purchase a trick in a magic store. And not the kind of mystical experience that organized religion can go on about. No, games are magic in the way that first kisses are magic, the way that finally arriving at a perfect solution to a difficult problem is magic, the way that conversation with close friends over good food is magic.

The magic at work in games is about finding hidden connections between things, in exploring the way that the universe of a game is structured. As all game players know, this kind of discovery makes for deeply profound experiences. How is it possible that the simple rules of chess and Go continue to evolve new strategies and styles of play, even after centuries and centuries of human study? How is it that the nations of the entire world, and even countries at war with each other—at war!—can come together to celebrate in the conflict of sport? How do computer and video games, seemingly so isolating, pierce our individual lives and bring us together in play?

To play a game is to realize and reconfigure these hidden connections—between units on a game board, between players in a match, between life inside the game and life outside—and in so doing, create new meaning. And if games are spaces where meaning is made, game designers are the meta-creators of meaning, those who architect the spaces of possibility where such discovery takes place.

Which is where this book comes in. You are reading these words because you are interested in not just playing games, but in making them. Take my word for it: *Game Design Workshop* is one of the very few books that can truly help you to make the games that you want to make. Those games bursting from your heart and from your imagination. The ones that keep you up at night demanding to be designed. Games that are brimming with potential for discovery, for meaning, for magic.

Game Design Workshop presents, with sharp intelligence and an eye for the importance of the design process, tried-and-true strategies for thinking about and creating games. More than just fancy notions about how games work, *Game Design Workshop* is a treasury of methods for putting game design theories into practice. The authors of *Game Design Workshop* have real experience making games, teaching game designers, and writing about game design. And I can honestly say that they have personally taught me a great deal. In the ambition of its scope and the value of its insights, you hold in your hands a very unique text.

Why do we need a book like *Game Design Workshop*? Because despite the fact that games are so very ancient, are part of every society, and are increasingly important in people's lives, we hardly know anything about them. We are still learning. What makes games tick? How do we create them? How do they fit into culture at large? The explosion

of computer and video games in recent decades has multiplied the complexity and the stakes of such questions. For better or worse, questions like these don't have simple answers. And *Game Design Workshop* won't give them to you. But it can help you figure out how to explore them on your own, through the games you design.

We are living through the rebirth of an ancient form of human culture. Just as the nineteenth century ushered in mechanical invention, and the twentieth century was the age of information, the twenty-first will be a century of play. As game designers, we will be the architects, the storytellers, and the party hosts of this playful new world. What a wonderful and weighty responsibility we have. To bring meaning to the world. To bring magic into the world. To make great games. And to set the world on fire through play.

Are you with me?

Eric Zimmerman
New York City, October 2007

Acknowledgments

The authors wish to thank the many game designers, producers, executives, and educators who have provided invaluable ideas, information, and insights during the writing of this book and the original edition. These talented individuals include:

Steve Ackrich, Activision
Phil Adams, Interplay
Graeme Bayless, Kush Games
Ranjit Bhatnagar, Gamelab
Seamus Blackley, CAA
Jonathan Blow
Chip Blundell, Eidos
Ian Bogost, Persuasive Games
Chris Brandkamp, Cyan
Brenda Brathwaite, Savannah College of Art and Design
Jeff Chen, Activision
Jenova Chen, thatgamecompany
Stan Chow, EA Japan
Doug Church, Electronic Arts
Dino Citraro, Periscopic
Don Daglow, Stormfront Studios
Elizabeth Daley, USC School of Cinematic Arts
Rob Daviau, Hasbro Games
Bernie DeKoven
Jason Della Rocca, IGDA
Dallas Dickinson, Sony Online Entertainment
Neil Dufine
Peter Duke, Duke Media
Troy Dunniway, Brash Entertainment
Greg Ecker
Glenn Entis, Electronic Arts
James Ernest, Cheapass Games
Noah Falstein, The Inspiracy
Dan Fiden, Electronic Arts
Matt Firor, Zenimax Online Studios
Scott Fisher, USC School of Cinematic Arts
Nick Fortugno, Rebel Monkey
Tom Frisina, Electronic Arts
Bill Fulton, Microsoft Game Studios
Richard Garfield, Wizards of the Coast
John Garrett, LucasArts
Chaim Gingold, Electronic Arts
Greg Glass
Susan Gold, IGDA Education SIG
Bing Gordon, Electronic Arts
Sheri Graner Ray, Women in Games International
Bob Greenberg, R/GA Interactive
Michael Gresh
Gary Gygax
Justin Hall, GameLayers
Brian Hersch, Hersch and Company
Richard Hilleman, Electronic Arts
Kenn Hoekstra, Pi Studios
Leslie Hollingshead, Vivendi Universal Games
Josh Holmes, Propaganda Games
Robin Hunicke, Electronic Arts
Steve Jackson, Steve Jackson Games
Matt Kassan, Atari
Kevin Keeker, Microsoft Games User Research
Heather Kelley
Scott Kim
Naomi Kokubo, Rocketon
Vincent Lacava, Pop and Co.
Lorne Lanning, Oddworld Inhabitants
Frank Lantz, area/code

Nicole Lazzaro, XEODesign
Marc LeBlanc, Mind Control Software
Tim Lee, Whyville
Nick Lefevre, Konami of America
Richard Lemarchand, Naughty Dog
Ethan Levy, PlayFirst
Rich Liebowitz, Union Entertainment
Starr Long, NC Soft
Sus Lundgren, PLAY Research Group
Michael Mateas, University of California, Santa Cruz
American McGee, Spicy Horse Games
Jane McGonigal, The Institute for the Future
Jordan Mechner
Nikita Mikros, Tiny Mantis Entertainment
Scott Miller, 3D Realms
Peter Molyneaux, Lionhead Studios
Alan R. Moon
Minori Murakami, Namco
Janet Murray, Georgia Institute of Technology
Ray Muzyka, BioWare
Dan Orzulak, Electronic Arts
Trent Oster, BioWare
Rob Pardo, Blizzard Entertainment
Celia Pearce, Georgia Institute of Technology
David Perry, Gameconsultants.com
Sandy Petersen, Ensemble Studios
Chris Plummer, Electronic Arts
Rhy-Ming Poon, Activision
Kim Rees, Periscopic
Stephanie Reimann, Nintendo
Neal Robison, Vivendi Universal Games
John Rocco
Bill Roper, Flagship Studios
Kate Ross, Wizards of the Coast
Rob Roth
Jason Rubin
Chris Rubyor, Petroglyph
Susana Ruiz
Katie Salen, Gamelab Institute of Play
Kellee Santiago, thatgamecompany
Jesse Schell, Carnegie Mellon University
Carl Schnurr, Activision
Steve Seabolt, Electronic Arts
Bruce C. Shelley, Ensemble Studios
Tom Sloper, Sloperama Productions
Warren Spector, Junction Point Studios
Jen Stein, USC School of Cinematic Arts
Michael Sweet, AudioBrain
Steve Swink, Flashbang Studios
Chris Taylor, Gas Powered Games
Brian Tinsman, Wizards of the Coast
Eric Todd, Electronic Arts
Kurosh ValaNejad, USC EA Game Innovation Lab
Jim Vessella, Electronic Arts
Jesse Vigil, Psychic Bunny
Steve Weiss, Sony Online Entertainment
Jay Wilbur, Epic Games
Dennis Wixon, Microsoft Games User Research
Will Wright, Electronic Arts
Richard Wyckoff, Pandemic Studios
Eric Zimmerman, Gamelab

We would also like to thank our editors and agents at Elsevier, Morgan Kaufmann, CMP, and Waterside Productions:

Dorothy Cox, CMP Books
Danielle Jatlow, Waterside Productions
Georgia Kennedy, Elsevier
Laura Lewin, Elsevier
Carol McClendon, Waterside Productions
Jamil Moledina, CMP Books
Dawnmarie Simpson, Elsevier
Paul Temme, Elsevier

And, of course, all of our students at the University of Southern California.

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Introduction

One of the most difficult tasks people can perform, however much others may despise it, is the invention of good games.

— C.G. Jung

Games are an integral part of all known human cultures. Digital games, in all their various formats and genres, are just a new expression of this ancient method of social interaction. Creating a good game, as noted in the Jung quote above, is a challenging task, one that requires a playful approach but a systematic solution. Part engineer, part entertainer, part mathematician, and part social director, the role of the game designer is to craft a set of rules within which there are means and motivation to play. Whether we are talking about folk games, board games, arcade games, or massively multiplayer online games, the art of game design has always been to create that elusive combination of challenge, competition, and interaction that players just call "fun."

The cultural impact of digital games has grown to rival television and films as the industry has matured over the past three decades. Game industry revenues have been growing at a double-digit rate for years and have recently eclipsed the domestic box office revenues of the film industry, reaching 12.5 billion dollars in 2007. According to reports in *Time Magazine* and *The LA Times*, 90% of U.S. households with children have rented or owned a video or computer game, and young people in the United States spend an average of 20 minutes per day playing video games. This makes digital games the second most popular form of entertainment after television.

As sales of games have increased, interest in game design as a career path has also escalated. Similar to the explosion of interest in screenwriting and directing that accompanied the growth of the film and television industries, creative thinkers today are turning to games as a new form of expression. Degree programs in game design are now available in major universities all over the world in response to student demand. The International Game Developers Association, in recognition of the overwhelming interest in learning to create games, has established an Education SIG to help educators create a curriculum that reflects the real-world process of professional game designers. On their website, the IGDA lists over 200 programs that offer game design courses or degrees in North America alone. Furthermore, *Game Developer* magazine puts out an annual career guide bonus issue to connect the study of game development to the practice of it.

In addition to our experience designing games for companies such as Disney, Sony, Sega, and Microsoft, the authors of this book have spent twelve years teaching the art of game design to students from a variety of different backgrounds and experience levels and have established a game design curriculum for the interactive media degrees at the USC School of Cinematic Arts. In this time, we have found that there are patterns in the way that beginning designers grasp the structural elements of games, common traps that they fall into, and certain types of exercises that can help them learn to make better games. This book encapsulates the experience we have gained by working with our students to design, prototype, and playtest hundreds of original game concepts.

Our students have gone on to jobs in all areas of the game industry, including game design, producing, programming, visual design, marketing, and quality assurance. Several of them have gone on to become prominent independent game designers, such as the team at thatgamecompany, which developed the hit downloadable title flOw from a student research project created at USC. The method we present here has proven to be successful over and over again. Whatever your background, your technical skills, your reasons for wanting to design games, our goal with this book is to enable you to design games that engage and delight your players.

Our approach is exercise driven and extremely nontechnical. This may surprise you, but we do not recommend implementing your designs digitally right away. The complexities of software development often hamper a designer's ability to see the structural elements of their system clearly. The exercises contained in this book require no programming expertise or visual art skills and so release you from the intricacies of digital game production while allowing you to learn what works and what does not work in your game system. Additionally, these exercises will teach you the most important skill in the game design: the process of prototyping, playtesting, and revising your system based on player feedback.

There are three basic steps to our approach:

Step 1

Start with an understanding of how games work. Learn about rules, procedures, objectives, etc. What is a game? What makes a game compelling to play? Part I of this book covers these game design fundamentals.

Step 2

Learn to conceptualize, prototype, and playtest your original games. Create rough physical or digital prototypes of your designs that allow you to separate the essential system elements from the complexities of full production. Put your playable prototype in the hands of players and conduct playtests that generate useful, actionable feedback. Use that feedback to revise and perfect your game's design. Part II, starting on page 147, covers these important design skills.

Step 3

Understand the industry and the place of the game designer in it. The first two steps give you the foundation of knowledge to be a literate and capable game designer. From there you can pursue the specialized skills used in the game industry. For example, you can pursue producing, programming, art, or marketing. You might become a lead game designer or perhaps one day run a whole company. Part III, starting on page 347 of this book, covers the place of the game designer on a design team and in the industry.

The book is full of exercises intended to get you working on game design problems and creating your own designs. When you reach the end, you will have prototyped and playtested many games, and you will have at least one original playable project of your own. We emphasize the importance of doing these exercises because the only way to really become a game designer is to make games, not just play them or read about them. If you think of this book as a tool to lead you through the process of design, and not just a text to read, you will find the experience much more valuable.

So if you are ready to get started, it's your turn now. Best of luck!